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RESEARCH ARTICLE

A Hybrid Approach for Predicting Corporate Financial Risk: Integrating SMOTE-ENN and NGBoost

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ABSTRACT The financial condition of an enterprise is a critical factor in determining its viability and long-term prospects, and it also directly affects the interests of stakeholders such as investors and creditors. So extra attention is needed to evaluate and forecast the financial risk of an enterprise. This paper proposes a novel enterprise financial risk evaluation and prediction model integrating Synthetic Minority Over-sampling and Edited Nearest Neighbors (SMOTE-ENN) and Natural Gradient Boosting (NGBoost). The CRITIC-TOPSIS method is first employed to conduct a comprehensive evaluation of the financial risk of the listed companies based on their annual financial statements. Subsequently, the obtained evaluation results are subjected to a clustering analysis to categorize the companies into different levels of financial risk. To address the issue of imbalanced samples, the study introduces the SMOTE-ENN combined sampling method. This technique is applied to mitigate the imbalance in the number of samples across different categories. Lastly, NGBoost is utilized to develop a prediction model for corporate financial risk based on the clustered data. The analysis reveals that companies with low levels of financial risk typically exhibit a balanced financial profile, robust operational performance, and substantial growth potential. Conversely, companies with high levels of financial risk are characterized by weak profitability and debt-paying ability. The experimental findings show that the model has a 97.18% accuracy rate, thereby suggesting that the model is both reasonable and practicable.

INDEX TERMS Financial risk, listed company, natural gradient boosting, SMOTE-ENN.

I. INTRODUCTION

Enterprises now face fiercer competition as a result of the social economy's quick expansion and the expanding integration of the global economy [1]. The effective management and mitigation of financial risk are intricately linked to the sustainable growth of enterprises, prompting an increasing number of corporations to augment their investments in financial risk evaluation and prediction. The enterprise financial risk evaluation system serves as a vital management tool for guiding and controlling modern corporations [2]. Moreover, it constitutes a significant theoretical foundation in management [3]. Stakeholders, including shareholders, creditors, employees, and government management, are all concerned about the financial risk level of listed firms [4]. Consequently, the scientific and equitable evaluation of financial risks in listed companies has emerged as a pivotal topic in the realm of financial research. Simultaneously, investigating methods to accurately predict financial risks in enterprises provides practical guidance for enhancing the operational conditions and efficiency of listed companies.

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Since the 1930s, scholars have dedicated their efforts to researching early warning systems for financial risk. Over time, a more comprehensive theoretical framework has been established, evolving from the initial single-factor consideration to the comprehensive consideration of multiple variables and indicators. Eventually, this progression led to the utilization of big data models for simulation analysis and verification. Broadly speaking, the development stages of financial risk early warning can be categorized into three major phases: univariate models, multivariate models, and machine learning algorithm models.

In the 1930s, Fitzpatrick [5] introduced the concept of early warning systems for financial risk. He proposed conducting financial forecasting studies on insolvent companies using Single Variable Analysis (SVA) based on their financial statements. Fitzpatrick utilized individual financial indicators, such as net equity margin and equity ratio, as reliable early warning signals for a company's financial condition. Although univariate financial forecasting models offer simplicity, they often present a less comprehensive picture of a company's financial situation [6]. Consequently, many scholars and entrepreneurs have shifted their focus towards studying and employing multivariate variables as early warning indicators. Beaver [7] conducted a study using 10 years of financial data from 158 companies and extracted 30 financial indicators to construct a risk warning model. Among the indicators, total debt to total assets, cash flow, and net income exhibited the highest predictive ability for financial risk. Additionally, Altman [8], an American scholar, developed a five-variable Z-score model by observing bankrupt and non-bankrupt production enterprises in the US. Altman's method employed Multiple Variable Analysis, laying the theoretical foundation for multivariate financial warning models. Altman et al. [9] et al. further enhanced the Z-score model by introducing the Zeta model, leading to improved prediction accuracy. The Zeta model remains the most widely used multivariate linear discriminant model globally. However, it is worth noting that multivariate analysis methods assume independent normal distributions for predictor variables, which may introduce bias in their predictions. To address this issue, Martin [10] were the first to introduce Logistic Regression(LR) methods to early warning systems for corporate financial risk, effectively addressing confounding factors encountered in multivariate analysis.

In response to the continuous advancements in modern information technology, particularly computer technology and artificial intelligence, scholars have begun incorporating machine learning algorithms into enterprise financial risk early warning systems. Prominent algorithms such as Backpropagation (BP) neural network, Random Forest (RF), and Support Vector Machine (SVM) have gained widespread usage in financial risk forecasting. To estimate the line risk of businesses, Sharma [11] created a Random Forest Model and evaluated its effectiveness against the conventional Logistic Regression approach. According to their research, RF outperformed LR in credit risk predicting. In order to forecast corporate misconduct in the construction business, Wang et al. [12] propose s a RF model that uses characteristics linked with corporate misconduct to rank their relevance. Shen et al. [13] introduced a Random Forest model combined with the ANS-REA algorithm, enabling multiple forecasts from an unbalanced data stream and providing an effective solution for financial distress forecasting. Support Vector Machine was used by AJHM and BYCL [14] to forecast firm bankruptcy. SVM outperformed the other techniques, according to their comparison with Multiple Discriminant Analysis (MDA), Logistic Regression, and three-layer fully connected BP neural networks. rbs-SVM is a hybrid integration technique that Wang and Ma [15] devised, which combined bagging, random subspaces, and Support Vector Machines to enhance model performance. Sun et al. 16 applied the ADE-SVM approach to combine the inflow of new data batches for corporate financial risk forecasting with a time course. This approach demonstrated superior performance compared to three traditional dynamic modeling approaches. Lyu [17] conducted an empirical analysis utilizing LR and BP neural network models for the purpose of examining and contrasting financial data. The findings of this study highlighted a noteworthy divergence between the two models. The results unequivocally revealed that the BP neural network model consistently exhibited superior predictive capabilities when it came to discerning early warning signals in corporate financial contexts. Using BP neural networks and a sizable dataset, Du et al. [18] created an Internet credit risk early warning model. To enhance the model's performance and boost early warning precision, they included a genetic algorithm (GA). Li et al. [19] optimized BP neural network, and achieved a correct prediction rate of over 80% for financial distress among selected companies. Cao and Wang [20] conducted a stock price prediction study using both financial indicator data and trading indicator data as input variables. They employed three BP neural network algorithms and discovered that the PCA and BP neural network-based stock price prediction models worked well. Li et al. [21]presents an innovative framework comprising a novel cluster quality evaluation criterion and an efficient solver. This approach is tailored for financial mining applications, addressing the challenges of interpreting intricate data distributions and detecting anomalies within financial datasets. Li's methodology stands out for its ability to significantly enhance the precision and cohesion of identified clusters by refining their centroids and scopes. Zhang et al. [22] introduce an innovative profit-driven prediction model, strategically engineered to maximize financial gains. This model represents a pivotal development in the field, offering platforms and lenders a potent tool for optimizing their decision-making processes. By focusing on profit maximization, it not only enhances the efficiency of identifying potential defaulters but also provides valuable insights for stakeholders in the lending industry. Kou et al. [23] have innovatively tailored a bankruptcy prediction model to cater specifically to Small and Medium-sized Enterprises (SMEs). Their model's uniqueness lies in its utilization of transactional data and payment network-based variables, eliminating the need for conventional financial accounting data. Additionally, they introduced a two-stage multiobjective feature-selection approach, streamlining the feature selection process. Notably, their research demonstrates that this model achieves comparable classification performance to traditional methods, all the while substantially reducing the requisite number of features for analysis.

In the area of predicting corporate financial risk, Ensemble Learning methods have demonstrated superior generalization, robustness, and interpretability compared to traditional machine learning approaches. As a result, they are more suitable for real-world production environments. Sun et al. [24] introduced two DFDP algorithms that integrated time-weighted techniques and Adaboost Support Vector Machines to achieve dynamic forecasting of enterprise financial distress. Du et al. [25] constructed a model by combining CUS with GBDT and employed XGBoost as a heterogeneous classifier. The integrated model exhibited improved generalization performance. Nobre and Neves [26] suggested a financial expert system that incorporated PCA, DWT, XGBoost, and MOO-GA. This integration aimed to achieve a balance between moderate risk and high return. For the purpose of predicting P2P online loan defaults, Ma et al. [27] examined the efficacy of the LightGBM and XGBoost model. According to the study, the LightGBM model produced the most superior prediction results.

With the proliferation of deep learning techniques, researchers have embarked on applying this methodology to finance-related studies [28]. Luo et al.'s research [29] conducted a rigorous comparison of Deep Learning methods, specifically DBN, with traditional models like LR, Multilayer Perceptron, and SVM. The results unequivocally favored DBN, as it exhibited unparalleled classification performance, surpassing all other models in the study. Mai et al. [30] introduced a deep learning model specifically tailored for corporate bankruptcy prediction, leveraging textual disclosures as a primary data source. The model showcased remarkable predictive capabilities when employing textual information from disclosure documents. Hosaka [31] employed a unique approach by converting financial ratios into grayscale images and subsequently using Convolutional Neural Network (CNN) to make predictions about corporate bankruptcy. The trained network demonstrated superior performance in bankruptcy prediction tasks. In a seminal contribution by Craja et al. [32], a novel approach was introduced for the detection of statement fraud within company annual reports. This innovative method seamlessly integrated financial ratios with the critical aspect of management commentary, with a special emphasis on the Management Discussion and Analysis (MD&A) section. Notably, the study revealed that the implementation of a Hierarchical Attention Network (HAN) for extracting textual features from the MD&A section exhibited substantial potential in enhancing the accuracy of fraudulent activity detection.

The current research on enterprise financial risk early warning reveals certain challenges that need to be addressed. (1) The classification of enterprise risk level is limited to a binary categorization based solely on whether the "ST" treatment is present or not. This approach results in an imbalanced sample size [33], [34], [35]. Consequently, it fails to provide an effective and timely warning system. (2) The utilization of Deep Learning models in enterprise risk management necessitates a large number of samples to establish robust mapping relationships and requires substantial computational power. This increased reliance on resources adds to the overall cost of implementing Enterprise Risk Management (ERM). In contrast, Ensemble Learning methods exhibit less dependence on sample size and offer higher accuracy. (3) The majority of financial early warning studies primarily concentrate on assessing or predicting the financial situation of a business. Few research has created an extensive early warning system that combines intelligent evaluation and intelligent forecasting.

To tackle these issues, this study presents an innovative solution, integrating SMOTE-ENN and NGBoost, to formulate an enterprise financial risk evaluation and prediction model. The CIRTIC-TOPSIS method is utilized for comprehensive financial status evaluation of A-share businesses listed on the Shenzhen and Shanghai Stock Exchanges. The resulting scores are then clustered using K-means to determine the financial risk levels of these companies. NGBoost, an Ensemble Learning algorithm known for its exceptional performance, is employed in this study. Notably, there is a lack of examples showcasing the application of the NGBoost algorithm specifically for corporate financial risk prediction. Therefore, the NGBoost model is trained to intelligently predict corporate financial risk levels. To overcome the limitations of previous studies, which often relied on single-classification approaches for financial risk and faced challenges in obtaining labeled data, this research adopts a combination of supervised learning and unsupervised learning. Moreover, considering that companies with minimal financial risk and great financial risk constitute a small proportion of the total sample, while those with average risk levels represent the majority, the problem at hand can be viewed as a multi-classification issue with an imbalanced sample distribution. To enhance the model's accuracy, the hybrid sampling technique, SMOTE-ENN, is employed for data enhancement. The technology roadmap outlining the suggested methodology is illustrated in Fig. 1.

The work is set up as follows: Sect.II provides an elaborate account of the research methodology selection and the processing and analysis of sample data for the development of a novel indicator system. Sect.III provides a comprehensive account of the experiment's setup, including all relevant details. It further presents the results obtained from the experiment and conducts a thorough analysis of these outcomes.

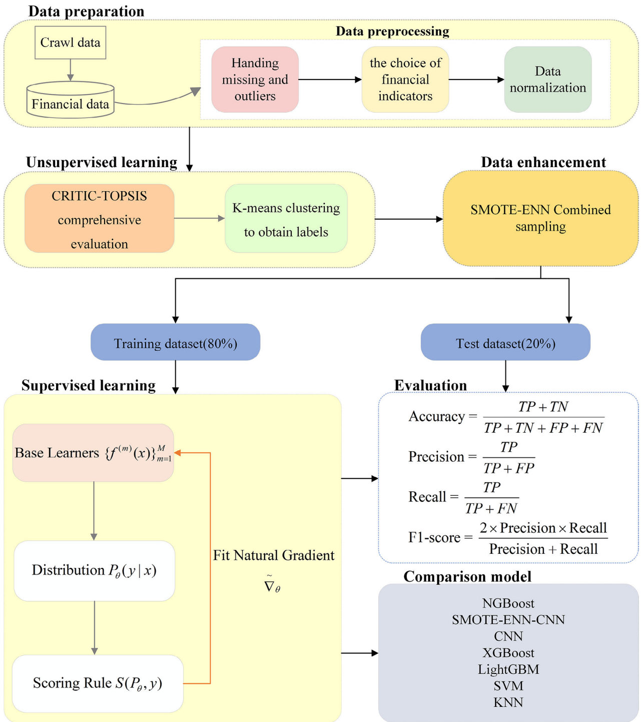

FIGURE 1. Technology roadmap of the financial risk prediction model.

Lastly, in Sect.IV, the paper draws its conclusions based on the findings and discussions presented throughout the work.

II. METHODOLOGY AND DATA

A. CRITIC-TOPSIS

The CRITIC (Criteria Importance Through Intercriteria Correlation) [36] utilized in the context of multi-criteria decision-making. This method delves into the intricate interdependencies among various criteria. By doing so, it strategically assigns weights to each criterion, leading to a more refined and precise representation of their respective importance in the decision-making process. In conjunction with the CRITIC-based weighting, the TOPSIS method is employed for ranking purposes, approximating the ideal solution. This technique establishes optimal and inferior solutions by constructing a weighted normalization matrix. The ranking is subsequently assessed by quantifying the object's distance to the optimal and inferior solutions. The following are the steps that make up the CRITIC-TOPSIS methods:

(1) Building the evaluation matrix. Assume there are n items to be evaluation and p standardized evaluation indications.

$$X = \begin{bmatrix} x_{11} & \cdots & x_{1p} \\ \vdots & x_{ij} & \vdots \\ x_{n1} & \cdots & x_{np} \end{bmatrix}$$

x_{ij} denotes the value of the j-th evaluation indicator for the i-th sample.

(2) Calculation of standard deviation for each indicator.

$$\begin{cases} \bar{x}_j = \frac{1}{n}\sum_{i=1}^{n} x_{ij} \\ S_j = \sqrt{\frac{\sum_{i=1}^{n}(x_{ij}-\bar{x}_j)^2}{n-1}} \end{cases} \tag{1}$$

S_j denotes the standard deviation of the j-th indicator

(3) Calculate the criterion conflict.

$$R_j = \sum_{i=1}^{p}(1-r_{ij}) \tag{2}$$

r_{ij} denotes the correlation coefficient between evaluation indicators i and j

(4) Calculate the information content.

$$C_j = S_j\sum_{i=1}^{p}(1-r_{ij}) = S_j \times R_j \tag{3}$$

(5) Calculate the objective weight of the j-th indicator w_j

$$w_j = \frac{C_j}{\sum_{j=1}^{p} C_j} \tag{4}$$

Following the above basic steps for CRITIC-based weighting, the use of the TOPSIS method is described in relation to the weights obtained w_j.

(6) Convergent decision matrix. On the basis of the original matrix R_x the negative indicators are homotopologized: $X'_{ij} = 1/X_{ij}$,to obtain the new decision matrix R'_x

$$R'_X = \begin{bmatrix} X'_{11} & \cdots & X'_{m1} \\ \vdots & X'_{ij} & \vdots \\ X'_{1n} & \cdots & X'_{mn} \end{bmatrix}$$

(7) Normalized decision matrix. To make data across indications more comparable, the TOPSIS technique also calls for normalizing the decision matrix. Usually, the matrix is normalized using the following formula:

$$X''_{ij} = X'_{ij} / \sqrt{\sum_{j=1}^{n} X'^{2}_{ij}} \tag{5}$$

The normalized decision matrix is R''_x.

$$R''_X = \begin{bmatrix} X''_{11} & \cdots & X''_{m1} \\ \vdots & X''_{ij} & \vdots \\ X''_{1n} & \cdots & X''_{mn} \end{bmatrix}$$

(8) The weights obtained by CRITIC are used to build the weighted normalization matrix. Each vector in the normalized decision matrix is multiplied with the indicator weight corresponding to that vector, and thus the weighted normalization matrix $V = [v_{ij}]$ is obtained.

$$V = \begin{bmatrix} v_{11} & \cdots & v_{m1} \\ \vdots & v_{ij} & \vdots \\ v_{1n} & \cdots & v_{mn} \end{bmatrix}$$

(9) Determine the positive and negative ideal solutions. The positive ideal solution V^+ consists of the positive indicator's highest value and the negative indicator's minimum value, and the negative ideal solution V^- is defined as the product of the minimum value of the positive indicator and the maximum amount of the negative indicator.

$$V^+ = \left\{ \left(\max_{1 \leq j \leq n} v_{ij} \right) \middle| j \in J, \left(\max_{1 \leq j \leq n} v_{ij} \right) \middle| j \in J \right\}$$
$$= \left\{ v_1^+, \cdots, v_j^+, \ldots, v_m^+ \right\}$$
$$V^+ = \left\{ \left(\min_{1 \leq j \leq n} v_{ij} \right) \middle| j \in J, \left(\max_{1 \leq j \leq n} v_{ij} \right) \middle| j \in J' \right\}$$
$$= \left\{ v_1^-, \cdots, v_j^-, \ldots, v_m^- \right\}$$

J is a positive indicator and J' is a negative indicator.

(10) Calculate the Euclidean distance. Separate calculations are made to determine the Euclidean distances between the assessed object and the ideal solutions, and then the evaluation ranking can be performed. The distance between the subject of the assessment and the ideal solution, both positive and negative, is provided by:

$$d_j^+ = \sqrt{\sum\nolimits_{i=1}^{m} (v_{ij} - v_j^+)^2} \tag{6}$$

$$d_j^- = \sqrt{\sum\nolimits_{i=1}^{m} (v_{ij} - v_j^-)^2} \tag{7}$$

(11) Calculate the Relative Closeness. Every object that needs to be assessed has its Euclidean distance calculated first, and then its relative closeness is utilized for the object's comprehensive evaluation value. The Relative Closeness is calculated using the following formula:

$$C_j = \frac{d_j^-}{d_j^+ + d_j^-} \; (j = 1, 2, \ldots n) \tag{8}$$

(12) Ranking the objects to be evaluated. The n item to be assessed are ranked according to the C_j. The larger the value of C_j, the farther the item is from the negative ideal solution and the closer it is to the positive ideal solution, the better the object is.

B. K-MEANS

Upon completion of the aforementioned scoring procedure, the manual grading of results introduces a subjective bias, thereby necessitating the utilization of an objective classification approach. We employ the K-means clustering [37]. K-means clustering's main goal is to divide samples into k different clusters, where samples within each cluster have the highest degree of similarity. The algorithm's sequential steps are outlined as follows:

(1) Choose k samples at random as the initial clustering centers.

(2) Calculate the Euclidean distance from the k clustering centers of a sample x_i with p features and assign x_i to the nearest clustering center:

$$dist\left(x_i, c_j\right) = \sqrt{\sum\nolimits_{k=1}^{p} (x_{ik} - c_{jk})^2} \tag{9}$$

$$c_i = argmin_{j=1,\ldots,k} dist\left(x_i, c_j\right) \tag{10}$$

(3) For each cluster, recalculate its cluster center:

$$c_j = \frac{1}{|S_j|} \sum\nolimits_{x_i \in S_j}^{n} x_i \tag{11}$$

(4) Till the clustering centers stop shifting or the maximum number of iterations has been achieved, repeat steps 2 and 3 as necessary.

In this paper, $k = 5$ was used to classify the financial risk of enterprises into five risk levels: "I", "II", "III", "IV", and "V" based on the CRITIC-TOPSIS scores of the selected samples.

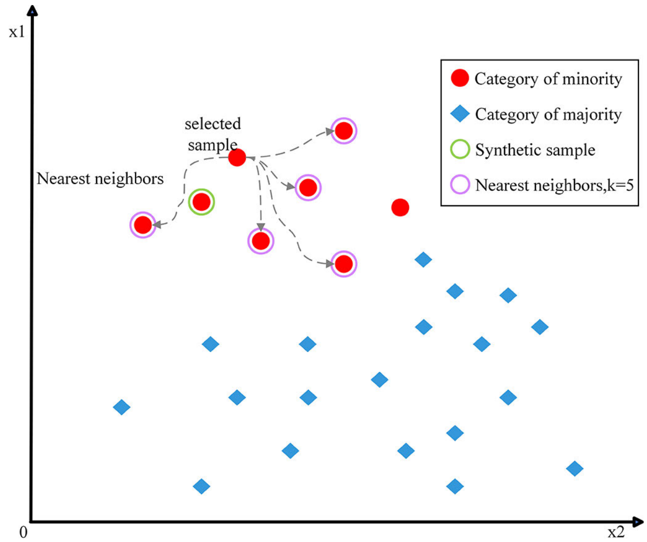

FIGURE 2. Smote flow char.

C. SMOTE-ENN

SMOTE-ENN, a data preprocessing technique that combines oversampling and undersampling, addresses the issue of imbalanced datasets. The mathematical underpinning of the SMOTE-ENN algorithm is as follows: initially, employing the conventional oversampling method known as SMOTE [38], illustrated in Fig. 2. The following are the particular steps:

(1) A minority class sample x is randomly selected.

(2) Find k (usually, $k = 5$) nearest minority class samples of x. Equation (8) is the formula for computing the Euclidean distance between samples.

(3) A sample y is randomly selected from the k nearest neighbors.

(4) For each feature i between x and y, a new sample x' is generated according to the following formula:

$$x_i' = x_i + rand\,(0, 1) \times (y_i - x_i) \tag{12}$$

whereas the SMOTE algorithm effectively addresses the issue of generating minority class samples, it does have certain limitations. The randomness principle employed in the nearest neighbor selection during the SMOTE process can potentially introduce noise, thereby posing challenges for model generalization. To mitigate the concerns, we introduce the EditedNearestNeighbours [39] as an improvement. The ENN algorithm has the following precise steps:

(1) Find the k nearest neighbors of each sample x from the majority class.

(2) Equation (9) is used to determine the Euclidean distance between each majority class sample x and its k nearest neighbors.

(3) For each majority class sample x, if the amount of minority class samples in its k nearest neighbors is smaller than the number of majority class samples, then x is marked as noise.

(4) Eliminate all samples from the majority class that have noise labels.

(5) The SMOTE-ENN method first creates a minority class of samples using the SMOTE, then uses the ENN to clean up the noise in the majority class of samples.

D. NATURAL GRADIENT BOOSTING

Natural Gradient Boosting(NGBoost), a modular algorithm for probabilistic regression [40], it employs natural gradients and multiparameter boosting to integrate any of the following: (1) Base learner; Any sklearn regressor can be used as the Base learner, as indicated by the Base parameter, the depth-3 regression tree is set as the default. (2) Parametric probability distribution; NGBoost can be used with a variety of distributions, such as LogNormal, Normal and Bernoulli. (3) Scoring rule; NGBoost supports the log score (LogScore, often known as negative log-likelihood) and CRPS (CRPScore), albeit each score may not be implemented for every distribution. NGBoost predicts the parameters of the conditional probability distribution of the model in functional form for the purpose of probabilistic prediction. The probability prediction with probability density of p_μ is generated by directly predicting the parameter μ, which provides a basis for the assessment of the credibility. the framework diagram of the NGBoost model is shown in Fig. 3.

Before explaining the principles of NGBoost, the proper scoring rule and divergence are first outlined. In terms of point prediction, the predictions are evaluated using a loss function in comparison to the observed data. In probabilistic regression, the proper scoring rule is employed to assess the agreement between the observed data and the estimated probability distribution.

A appropriate scoring rule S assigns a score $S(G, z)$ to an observation z(outcome) and a forecasted probability distribution G, ensuring that the true distribution of outcomes receives the highest expected score. A scoring rule S is a proper scoring rule in mathematical notation if and only if

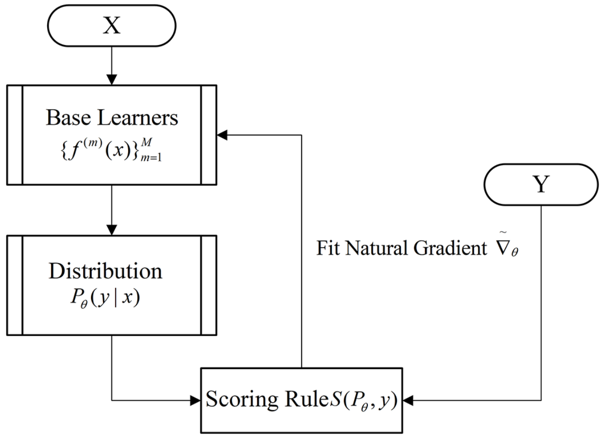

FIGURE 3. NGBoost offers modularity in selecting the scoring rule, distribution, and base learner.

it fulfills:

$$E_{z\sim g}\left(S\left(g,z\right)\right) \leq E_{z\sim g}\left(S\left(G,z\right)\right) \forall G, g \tag{13}$$

In this context, g denotes the true distribution of outcomes z, while G refers to any other distribution (e.g., a probabilistic forecast from a model).

The most popular proper scoring rule is the logarithmic score L, which, when minimized, gives the MLE:

$$L\left(\mu, z\right) = -\ln G_\mu\left(z\right) \tag{14}$$

Each proper scoring rule induces a divergence, which can function as a local distance metric within the distribution space. By definition, a proper scoring rule fulfills in (13), where the difference between the right-hand side and the left-hand side represents the divergence induced by that particular scoring rule. The Maximum Likelihood Estimation (MLE) scoring rule induces the Kullback-Leibler divergence (KL divergence or DKL) in the following manner.

$$\begin{aligned} D_L(g||G) &= E_{z\sim g}\left(S\left(G,z\right)\right) - E_{z\sim g}\left(S\left(g,z\right)\right) \\ &= E_{z\sim g}\left(\ln\frac{g\left(z\right)}{G\left(z\right)}\right) \end{aligned} \tag{15}$$

The detailed steps of NGBoost are as follows:

(1) Input dataset $U = \{(f_i, y_i)\}$, $i = 1,2,\ldots, m$. Boosting iterations M, Learning rate η, Probability distribution with parameter θ, Proper scoring rule S, Base learner f.

(2) The probabilistic prediction $y|x \sim P_\theta(x)$ of θ is obtained by (15).

$$argmax_\theta \sum\nolimits_{i=1}^{n} S\left(\theta, y_i\right) \tag{16}$$

(3) The first $\theta^{(0)}$ is obtained according to (15)

$$\theta = \theta^{(0)} - \eta \sum\nolimits_{i=1}^{M} \rho^{(m)} f^{(m)}\left(x\right) \tag{17}$$

(4) The algorithm iterates M times, and at each iteration, the algorithm gets the predicted parameter $\theta_i^{(}m-1)$ for each sample i, as well as the evaluation rule S and natural gradient

$g_i^(m)$ corresponding to $\theta_i^(m-1)$, $g_i^(m)$ and θ are of the same dimension.

(5) The predicted outputs are updated by scaling them with stage-specific scaling factors $\rho^(m)$ and a common learning rate η before updating $\theta_i^(m)$

(6) Output:

$$\left\{\rho^{(m)}, f^{(m)}\right\}_{m=1}^{M}$$

E. DATA AND DATA INDICATOR SYSTEM

1) DATA COLLECTION

In this study, a comprehensive dataset was harnessed, comprising financial information from a diverse spectrum of 4,502 A-share listed companies trading on the Shanghai Stock Exchange and Shenzhen Stock Exchange in China. These enterprises represent diverse industries, including manufacturing, banking, agriculture, forestry, and biotechnology. The financial data selected for analysis were obtained through web crawling from the eastmoney website (https://www.eastmoney.com/).

2) DATA INDICATOR SYSTEM

Financial indicators encompass various categories, such as solvency, operating capacity, growth capacity, profitability, per-share indicators, relative value indicators, ratio structure, risk level, and dividend distribution. The CRITIC-TOPSIS method relies on mathematical matrix operations, making the selection of appropriate indicator data crucial for an effective evaluation of financial risk. In this study, a comprehensive approach was taken by combining previous research [41], [42], considering the specific characteristics of Chinese listed companies [43], a careful screening and division process was applied to the initially gathered 46 indicators, excluding less indicative variables for enterprise performance prediction and eliminating similar variables. As a result, an index system comprising 27 indicators at 6 levels was constructed. These levels include Key Financial Indicators, Solvency, Growth Capability, Profitability, Ratio Structure, and Operating Capacity. Detailed composition information can be found in Table 1.

3) DATA PRE-PROCESSING

(1) During the data pre-processing phase, samples with a significant number of missing values were excluded from the dataset. As a result, a total of 4,502 initial data points were obtained, each containing 27 financial indicators. It is worth noting that among these indicators, there were 107 samples representing "ST" companies. To enhance the prediction model's performance, the selected set of 27 indicators encompasses both positive and negative indicators, ensuring a comprehensive representation of the financial landscape.

(2) Standardization of indicators. For the positive and negative indicators, they are standardized separately to eliminate the dimensionality problem of the indicators.

Positive indicators:

$$Y_{ij} = \frac{X_{ij} - \min(X_{1j}, \cdots, X_{mj})}{\max(X_{1j}, \cdots, X_{mj}) - \min(X_{1j}, \cdots, X_{mj})} \tag{18}$$

Negative indicators:

$$Y_{ij} = \frac{\min(X_{1j}, \cdots, X_{mj}) - X_{ij}}{\max(X_{1j}, \cdots, X_{mj}) - \min(X_{1j}, \cdots, X_{mj})} \tag{19}$$

Obtain the new indicator matrix:

$$R_y = \begin{bmatrix} Y_{11} & \cdots & Y_{m1} \\ \vdots & Y_{ij} & \vdots \\ Y_{1n} & \cdots & Y_{mn} \end{bmatrix}$$

III. RESULT AND DISCUSSION

A. RISK MEASUREMENT OF CRITIC-TOPSIS

The CRITIC-TOPSIS method presents notable advantages in the financial sector [44], [45], [46], [47]. In this study, we employ a sophisticated weighting methodology based on CRITIC to determine the significance of the selected indicators. This approach stands in stark contrast to the simplistic practice of assigning equal weights, offering a more scientifically informed way of allocating weights based on the disparities in parameter values among the indicators. By integrating CRITIC-based weighting with the Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS), this method provides an objective and unbiased calculation of indicator weights. This amalgamation not only yields a comprehensive and accurate evaluation of the importance of each indicator but also ensures that various types of indicators are treated equitably in the decision-making process. Through comprehensive ranking, it provides decision-makers with scientifically informed support for quantitative analysis-based decision-making, aiding effective evaluation and selection of optimal risk management strategies [48]. As the financial risk decreases, a company tends to approach the optimal solution and distance itself from the worst solution.

Table 2 presents a comprehensive depiction of the "ST" firms' distribution subsequent to their arrangement via the TOPSIS methodology. The presence of the "ST" marker typically indicates that these companies may be associated with certain financial risks or problems. Notably, an overwhelming majority, exceeding 80%, of the "ST" firms are situated within the lowermost quartile, with a conspicuous absence of "ST" firms within the uppermost quartile. This discernment posits the efficacy of the CRITIC-TOPSIS scoring model in its aptitude to discern divergences in financial risk among firms.

Table 3 displays the results produced by implementing the CRITIC-TOPSIS approach for the analysis of all data samples, including the distances from the positive and negative ideal solutions, as well as the scores. Based on the application of the CRITIC-TOPSIS method, *Sino Biological, Inc.* achieved the highest score of 0.464, and *Inner Mongolia Western Resources Co., Ltd* obtained the lowest score of 0.283 among all the selected samples. On average,

TABLE 1. Selected indicators.

	Indicator name	Variable	Nature of indicator
Key Financial Indicators	Basic Earnings per Share (EPS)	X_1	Positive
	Operating Revenue	X_2	Positive
	Net Non-Operating Income and Expenditure	X_3	Positive
	Net Increase in Cash and Cash Equivalents	X_4	Positive
	Weighted Average Return on Equity (ROE)	X_5	Positive
Solvency	Current Ratio	X_6	Positive
	Cash Ratio	X_7	Positive
	Interest Coverage Ratio	X_8	Positive
	Debt Asset Ratio	X_9	Negative
	Equity Ratio	X_{10}	Negative
Growth Capability	Growth Rate of Main Business Revenue	X_{11}	Positive
	Total Asset Growth Rate	X_{12}	Positive
Profitability	Return on Total Assets (ROTA)	X_{13}	Positive
	Main Operating profit margin	X_{14}	Positive
	Cost-Expense-Profit Margin	X_{15}	Positive
	Operating Profit Ratio	X_{16}	Positive
	Return on Assets (ROA)	X_{17}	Positive
	Net Profit Margin	X_{18}	Positive
	Net Profit Margin on Total Assets	X_{19}	Positive
	Net Profit Growth Rate	X_{20}	Positive
Ratio Structure	Fixed Asset Ratio	X_{21}	Positive
	Equity to Fixed Asset Ratio	X_{22}	Positive
Operating Capability	Accounts Receivable Turnover Ratio	X_{23}	Positive
	Inventory Turnover Ratio	X_{24}	Positive
	Total Asset Turnover Ratio	X_{25}	Positive
	Cash Flow to Sales Ratio	X_{26}	Positive

TABLE 2. Overall distribution of st enterprises.

	Number of "ST"	Percentage
The top 25%	0	0
The middle 50%	20	18.7%
The bottom 25%	87	81.3%

TABLE 3. Critic-Topsis method scores for selected listed companies.

Company name	Stock code	d^+	d^-	Score
Sino Biological,Inc.	301047	0.687	0.596	0.464
ACROBIOSYSTEMS Co., Ltd.	301080	0.69	0.591	0.462
YongYue Science&Technology Co., Ltd.	603879	0.707	0.584	0.452
Shenwu Energy Saving Co., Ltd.	000820	0.715	0.585	0.45
Wuchan Zhongda Group Co., Ltd.	600704	0.715	0.526	0.424
…	…	…	…	…
*ST Galaxy Biomedical Investment Co.,Ltd.	000806	0.833	0.368	0.306
*ST Shunliban Information Service Co.,Ltd	000606	0.838	0.366	0.304
*ST GI Technologies Group Co.,Ltd.	300309	0.826	0.354	0.3
*ST Shanghai New Culture Media Group Co.,Ltd.	300336	0.848	0.356	0.296
*ST Inner Mongolia Western Resources Co., Ltd	600139	0.884	0.349	0.283

the mean score across all samples was calculated as 0.368. Notably, 2273 samples received scores above the average, while 2229 samples obtained scores below the average.

Fig. 4 illustrates the indicator weights computed utilizing the CRITIC method. Evidently, Solvency and Profitability emerge as prominently weighted factors, commanding 27.11% and 19.77% respectively. This discernment underscores the imperative of upholding a steadfast financial framework and a commendable profit orientation in fostering enduring progress. The weightage assigned to Growth Capacity stands at 7.66%, underscoring the influence of an enterprise's potential for expansion on the broader financial risk landscape. Meanwhile, Operating Capability garners a 14.75% allocation within the overall weightage, accentuating the role of adept operational management and judicious resource allocation in mitigating the enterprise's financial risk profile. Notably, Ratio Structure commands the proportion at 12.76%.

Table 4 succinctly lists the basic financial metrics for the five healthcare industry companies, while Fig. 5 depicts a visual comparison of the solvency and profitability metrics for these five companies. Among them, *IMEIK Technology*

TABLE 4. Selected financial indicators for selected companies.

	Selected financial indicators	300896	688580	688677	*ST600532	*ST300273
Solvency	Current Ratio	21.3	13.81	12.66	1.67	1.68
	Cash Ratio	1827.43	1315.14	1063.21	0.4	108.59
	Debt-to-Asset Ratio	4.47	6.45	7.37	36.98	58.28
	Equity Ratio	3.55	6.78	7.72	54.48	102.91
Profitability	Main Operating profit margin	93.24	73.52	62.31	10.07	31.01
	ROA	19	12.33	14.61	-5.88	-4.31
	Net Profit Margin	66.12002545	41.28331068	37.83261806	-24.62419487	-52.95319648
	Net Profit Margin on Total Assets	19.34496301	11.16868068	14.65304615	-6.393359902	-6.26987385
Growth Capability	Growth Rate of Main Business Revenue	104.13	13.66	4.08	-83.72	-21.09
Operating Capability	Accounts Receivable Turnover Ratio	28.34	23.12	4.96	17.08	0.96

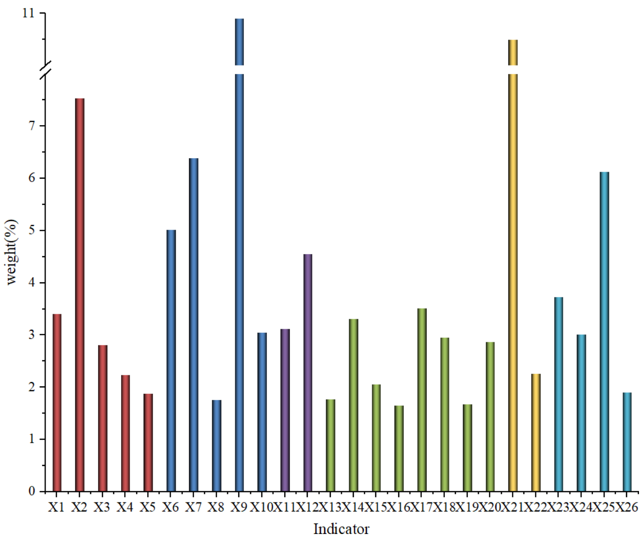

FIGURE 4. CRITIC weights for selected indicators.

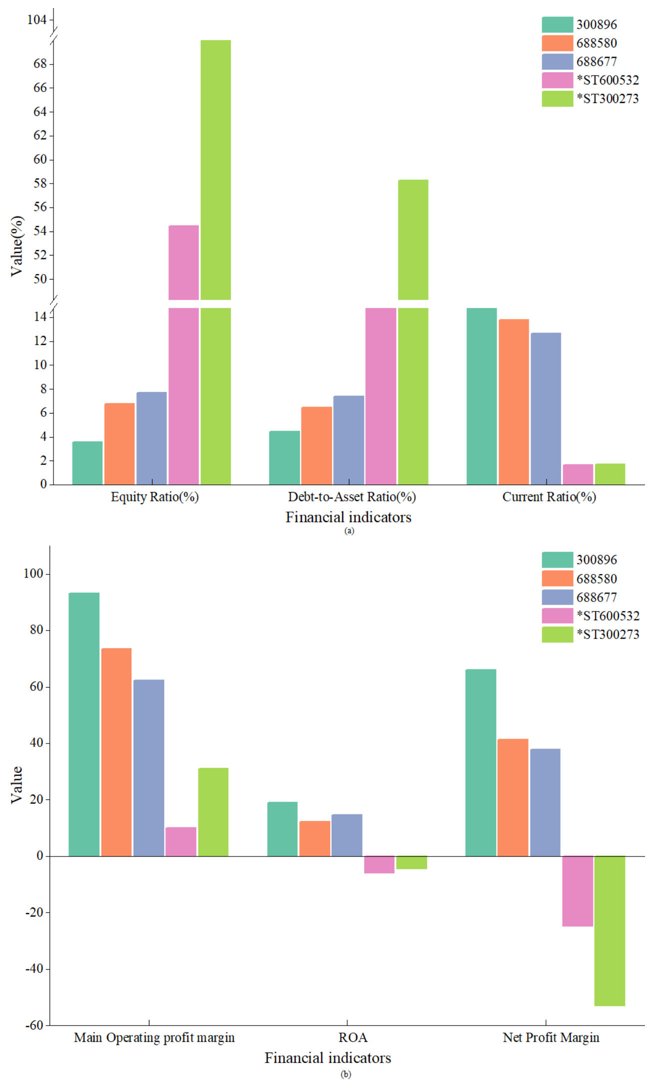

FIGURE 5. Comparison of financial indicators for selected companies.

Development Co., Ltd(300896), *Nanjing Vishee Medical Technology Co., Ltd.*(688580) and *Qingdao NovelBeam Technology Co., Ltd.* (688677) outperformed other companies in the field. However, it is noteworthy that ∗ST *Shanghai Topcare Medical Services Co.,Ltd.*(600532) and ∗ST *Zhuhai Hokai Medical Instruments Co., Ltd.*(300273) exhibited significantly lower scores compared to other companies.

When evaluating the financial solvency of enterprises, it becomes evident that specific indicators, including the Cash Ratio, Current Ratio and Debt Asset Ratio, wield considerable significance compared to their counterparts. These indicators play a pivotal role in assessing the financial risk associated with these enterprises. For instance, as depicted in Fig.5(a), the Debt Asset Ratio of high-performing enterprises significantly contrasts with that of their lower-ranking counterparts. Conversely, the Current Ratio and Cash Ratio of the former group significantly outpace those of the latter, culminating in the model's deduction of heightened solvency for the high-ranking enterprises. The Current Ratio stands as a crucial financial metric, gauging a company's ability to swiftly convert its current assets into cash, thus serving as a key indicator of its short-term solvency. Similarly, the Cash Ratio holds significance in evaluating short-term solvency, offering insights into an organization's immediate liquidity position. Generally, a higher Cash Ratio indicates stronger short-term solvency and greater capacity to manage short-term debt risks. The combination of a higher Debt Asset Ratio, lower Current Ratio, and lower Cash Ratio suggests a decline in the short-term solvency of these companies and difficulties in quickly liquidating their liquid assets, thereby exposing them to increased financial risks.

Amidst the metrics assessing corporate profitability, no individual metric commands a notably elevated weighting. However, the cumulative weighting surpasses the 19% threshold, signifying that the application of the CRITIC methodology provides a judicious evaluation of diverse dimensions across various profitability-related indicators. This approach, in turn, culminates in the provision of a more holistic depiction of profitability. The profit profiles of the five chosen enterprises are visually depicted in Fig. 5(b) and Table 4. It is evident that the two "ST" companies exhibited negative values for Net Profit Margin, ROA and Net Profit Margin on Total Assets, whereas the three highly rated companies demonstrated exceptional profitability within their respective industries. This indicates that the profitability of the two selected "ST" companies was notably weaker compared to the other companies. ROA, a significantly weighted profitability metric, encapsulates a company's profitability and the efficiency with which it utilizes its assets. When a company can generate higher profits with an equivalent amount of assets, it may be indicative of lower financial risk. Conversely, a diminished ROA might signal suboptimal asset utilization, potentially necessitating augmented investments to ensure business sustainability. The data analysis presented above affirms the model's deduction that highly ranked firms exhibit superior profitability in comparison to their lowly ranked counterparts.

Although the total weight of the indicators measuring growth capacity is lower than that of other categories, the weights of the two selected indicators are relatively high, and it can be seen from Table 4 that the three high-scoring enterprises are developing rapidly, and their growth capacity indicators are much higher than those of the low-scoring enterprises. From Fig. 4, we can see that the weights of the indicators related to operating ability are generally higher, and accordingly, the Accounts Receivable Turnover Ratio of the high-scoring enterprises is significantly better than that of the low-scoring enterprises. Augmented operational adeptness signifies an enterprise's dexterity in resourceful administration of operational processes and assets, concomitantly fostering enhanced efficiency, improved cash flow management, heightened market competitiveness, and a fortified financial standing.

To encapsulate, the high-ranking enterprises manifest superior performance across indicators bearing higher weightages, thereby indicating their adeptness in risk mitigation. This proficiency is rooted in their enhanced capacity to excel in critical domains encompassing strategic delineation, judicious allocation of resources, and astute debt administration. This substantiates the reasonable and dependable utilization of the CRITIC-TOPSIS method for assessing a company's financial risk.

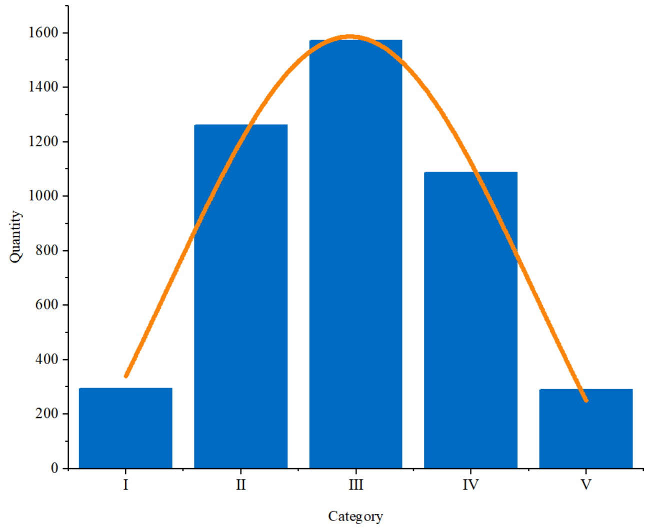

FIGURE 6. Clustering results.

B. RISK CLUSTERING OF K-MEANS

The K-Means clustering algorithm finds widespread application in the realm of finance [49], [50], [51]. In this study, we employ the K-Means algorithm to categorize selected enterprises. In order to distinguish the financial risk levels of different corporations, this study divides the financial risks of corporations into five categories: "I", "II", "III", "IV", "V", which mean very low financial risk, low financial risk, average financial risk, light financial warning, and heavy financial warning, respectively. Fig. 6 visually presents the distribution of financial risk among the five categories of companies, while Table 5 provides detailed clustering results for each category.

Table 5 illustrates the distribution of financial risk across different categories of enterprises. Among the analyzed samples, 6.55% of the enterprises were classified as having very low financial risk, 28.01% had low financial risk, 34.90% had average financial risk, 24.12% had a light warning, and 6.42% had a heavy warning. Notably, 84.1% of the ST enterprises fell into the "IV" or "V" categories, with no ST companies listed in the "I" or "II" categories. As depicted in Fig. 6, the distribution of samples within each category exhibits a comparable normal distribution pattern. Notably, the lowest count of firms is observed in the categories denoting very high and very low financial risk, while conversely, the highest concentration of firms falls within the bracket representing average financial risk.

Within this study, we meticulously curated a selection of indicators that wield substantial influence over the financial risk assessment process. Due to the non-normal distribution of the existing financial indicators, this study employs the Kruskal-Wallis test to examine the differences in the selected

TABLE 5. K-Means clustering of samples.

	Clustering result				The distribution of "ST" enterprises	
Risk level	Quantity	Percentage	Range of scores	Clustering center	Number of "ST"	Percentage
I	295	6.5526%	0.464~0.389	0.396364	0	0%
II	1261	28.0098%	0.388~0.375	0.380609	0	0%
III	1571	34.8956%	0.374~0.362	0.367838	17	15.9%
IV	1086	24.1226%	0.361~0.347	0.355133	20	18.7%
V	289	6.4194%	0.346~0.283	0.337593	70	65.4%

TABLE 6. Results of kruskal-wallis test.

	Cluster Category Median M(P25，P75)						
	Ⅰ(n=295)	Ⅱ(n=1261)	Ⅲ(n=1571)	Ⅳ(n=1086)	Ⅴ(n=289)	H	p
EPS	0.940(0.4,1.9)	0.540(0.2,1.1)	0.410(0.1,0.8)	0.200(0.0,0.6)	-0.250(-1.0,0.1)	725.107	<0.001
ROE	12.410(6.8,18.6)	9.600(5.0,15.4)	8.650(4.0,14.4)	5.735(0.6,11.2)	-9.070(-60.8,5.7)	497.895	<0.001
Current Ratio	9.340(6.9,13.3)	3.480(2.6,4.7)	1.750(1.4,2.2)	1.210(1.0,1.5)	0.895(0.7,1.1)	2777.947	<0.001
Cash Ratio	429.810(256.6,680.5)	118.220(68.4,198.4)	46.020(28.3,74.8)	28.155(18.8,40.8)	15.410(7.8,24.5)	2280.383	<0.001
Debt-to-Asset Ratio	9.260(6.9,11.7)	21.900(16.9,26.5)	41.260(35.8,47.0)	59.515(54.6,66.0)	83.060(77.4,89.7)	3773.108	<0.001
Equity Ratio	8.560(6.1,11.6)	24.380(17.6,32.3)	61.990(48.6,79.2)	133.345(107.4,175.7)	392.265(276.5,650.1)	3500.925	<0.001
Equity to Fixed Asset Ratio	947.010(496.3,2310.3)	480.900(287.3,1043.8)	341.270(199.1,680.1)	270.935(143.3,560.1)	189.630(60.5,834.1)	465.248	<0.001
Total Asset Growth Rate	20.390(5.5,116.0)	12.520(4.1,34.1)	12.620(3.8,26.6)	10.270(-0.1,24.9)	-0.270(-14.7,15.2)	224.057	<0.001
ROTA	7.570(4.5,12.2)	6.510(3.7,10.1)	4.690(2.2,7.4)	2.080(0.1,4.0)	-2.880(-14.4,0.7)	1225.303	<0.001
Main Operating profit margin	46.380(30.4,64.7)	34.280(23.5,48.5)	24.070(16.0,33.9)	19.175(12.0,28.3)	13.615(6.8,22.4)	890.633	<0.001
Cost-Expense-Profit Margin	36.330(21.0,63.4)	18.230(9.3,32.1)	9.670(4.1,18.1)	4.415(0.3,10.7)	-8.725(-43.0,3.4)	1200.541	<0.001
Operating Profit Ratio	25.790(15.6,37.4)	14.640(8.2,23.1)	8.530(3.8,14.6)	4.175(0.2,9.3)	-7.915(-45.5,3.2)	1133.817	<0.001
ROA	15.290(8.7,22.1)	17.270(10.8,24.9)	15.260(9.7,23.1)	10.460(5.2,17.6)	1.850(-7.4,6.7)	573.993	<0.001
Net Profit Margin	22.352(14.0,33.0)	12.845(7.2,20.2)	7.599(3.3,12.7)	3.612(0.3,7.9)	-9.921(-51.4,2.1)	1227.309	<0.001
Net Profit Margin on Total Assets	9.542(5.2,14.9)	7.170(3.9,11.2)	5.039(2.3,8.1)	2.203(0.1,4.4)	-2.936(-13.8,0.8)	1235.735	<0.001
Net Profit Growth Rate	17.903(-7.7,58.8)	12.901(-19.4,53.5)	14.508(-22.0,62.7)	12.005(-48.6,66.2)	-28.430(-239.8,39.4)	90.088	<0.001
Accounts Receivable Turnover Ratio	6.290(3.5,15.0)	5.610(3.2,11.1)	5.220(3.2,10.8)	4.945(2.8,9.7)	4.350(2.2,16.4)	24.637	<0.001
Cash Flow to Sales Ratio	0.190(0.1,0.3)	0.120(0.0,0.2)	0.070(0.0,0.1)	0.050(-0.0,0.1)	0.040(-0.1,0.1)	267.957	<0.001

indicators among categories. In fact, many financial datasets often exhibit certain non-normal distribution characteristics, such as skewness or fat tails [52], [53], [54], [55]. This is attributed to various influencing factors on financial data, including market volatility, economic cycles, human behavior, and more. Detailed outcomes of the Kruskal-Wallis test analyses have been meticulously tabulated in Table 6. It can be seen that the p-values of the selected indicators are all less than 0.001, indicating that there are significant differences in these indicators between different categories of enterprises. Table 6 further furnishes the median, 25th percentile and 75th percentile statistics for the chosen indicators. Notably, the median of positive indicators, such as ROTA, exhibit a gradual decrement as one traverses from Category "I" to Category "V". Conversely, the median values of negative indicators, such as Equity Ratio, increment progressively from Category "I" to Category "V". Subsequently, this study employed Dunn's test for further investigation to determine which specific groups or categories differ significantly from each other. The results of the Dunn's test are presented in Table 7. From Table 7, it can be observed that the majority of p-values are less than 0.001, indicating significant differences among most groups. However, it should be noted that the sample sizes for category "I" and "V" are only 200, which may lead to the non-significance of differences in a few pairwise group comparisons. Tables 6 and 7 collectively demonstrate that the results obtained from K-Means clustering effectively differentiate between different categories.

TABLE 7. Results of Dunn's test.

	Category I	Category J	Median I	Median J	Difference I-J	*p*
EPS	I	III	0.94	0.41	0.53	<0.001
	I	II	0.94	0.54	0.4	<0.001
	I	V	0.94	-0.25	1.19	<0.001
	I	IV	0.94	0.2	0.74	<0.001
	III	II	0.41	0.54	-0.13	<0.001
	III	V	0.41	-0.25	0.66	<0.001
	III	IV	0.41	0.2	0.21	<0.001
	II	V	0.54	-0.25	0.79	<0.001
	II	IV	0.54	0.2	0.34	<0.001
	V	IV	-0.25	0.2	-0.45	<0.001
ROE	I	II	12.41	9.6	2.81	<0.001
	I	III	12.41	8.65	3.76	<0.001
	I	IV	12.41	5.735	6.675	<0.001
	I	V	12.41	-9.07	21.48	<0.001
	II	IV	9.6	5.735	3.865	<0.001
	II	V	9.6	-9.07	18.67	<0.001
	III	II	8.65	9.6	-0.95	<0.001
	III	IV	8.65	5.735	2.915	<0.001
	III	V	8.65	-9.07	17.72	<0.001
	V	IV	-9.07	5.735	-14.805	<0.001
Current Ratio	I	II	9.34	3.48	5.86	<0.001
	I	III	9.34	1.75	7.59	<0.001
	I	IV	9.34	1.21	8.13	<0.001
	I	V	9.34	0.895	8.445	<0.001
	II	IV	3.48	1.21	2.27	<0.001
	II	V	3.48	0.895	2.585	<0.001
	III	II	1.75	3.48	-1.73	<0.001
	III	IV	1.75	1.21	0.54	<0.001
	III	V	1.75	0.895	0.855	<0.001
	V	IV	0.895	1.21	-0.315	<0.001
Cash Ratio	I	II	429.81	118.22	311.59	<0.001
	I	III	429.81	46.02	383.79	<0.001
	I	IV	429.81	28.155	401.655	<0.001
	I	V	429.81	15.41	414.4	<0.001
	II	IV	118.22	28.155	90.065	<0.001
	II	V	118.22	15.41	102.81	<0.001
	III	II	46.02	118.22	-72.2	<0.001
	III	IV	46.02	28.155	17.865	<0.001
	III	V	46.02	15.41	30.61	<0.001
	V	IV	15.41	28.155	-12.745	<0.001
Debt-to-Asset Ratio	I	II	9.26	21.9	-12.64	<0.001
	I	III	9.26	41.26	-32	<0.001
	I	IV	9.26	59.515	-50.255	<0.001
	I	V	9.26	83.06	-73.8	<0.001
	II	IV	21.9	59.515	-37.615	<0.001
	II	V	21.9	83.06	-61.16	<0.001
	III	II	41.26	21.9	19.36	<0.001
	III	IV	41.26	59.515	-18.255	<0.001
	III	V	41.26	83.06	-41.8	<0.001
	V	IV	83.06	59.515	23.545	<0.001
Equity Ratio	I	II	8.56	24.38	-15.82	<0.001
	I	III	8.56	61.99	-53.43	<0.001
	I	IV	8.56	133.345	-124.785	<0.001
	I	V	8.56	392.265	-383.705	<0.001
	II	IV	24.38	133.345	-108.965	<0.001
	II	V	24.38	392.265	-367.885	<0.001
	III	II	61.99	24.38	37.61	<0.001
	III	IV	61.99	133.345	-71.355	<0.001
	III	V	61.99	392.265	-330.275	<0.001
	V	IV	392.265	133.345	258.92	<0.001

TABLE 7. ***(Continued.)*** **Results of Dunn's test.**

Equity to Fixed Asset Ratio	I	II	947.01	480.9	466.11	<0.001
	I	III	947.01	341.27	605.74	<0.001
	I	IV	947.01	270.935	676.075	<0.001
	I	V	947.01	189.63	757.38	<0.001
	II	IV	480.9	270.935	209.965	<0.001
	II	V	480.9	189.63	291.27	<0.001
	III	II	341.27	480.9	-139.63	<0.001
	III	IV	341.27	270.935	70.335	<0.001
	III	V	341.27	189.63	151.64	<0.001
	V	IV	189.63	270.935	-81.305	0.049
Total Asset Growth Rate	I	II	20.39	12.52	7.87	<0.001
	I	III	20.39	12.62	7.77	<0.001
	I	IV	20.39	10.27	10.12	<0.001
	I	V	20.39	-0.27	20.66	<0.001
	II	IV	12.52	10.27	2.25	<0.001
	II	V	12.52	-0.27	12.79	<0.001
	III	II	12.62	12.52	0.1	0.039
	III	IV	12.62	10.27	2.35	<0.001
	III	V	12.62	-0.27	12.89	<0.001
	V	IV	-0.27	10.27	-10.54	<0.001
ROTA	I	II	7.57	6.51	1.06	0.001
	I	III	7.57	4.69	2.88	<0.001
	I	IV	7.57	2.08	5.49	<0.001
	I	V	7.57	-2.88	10.45	<0.001
	II	IV	6.51	2.08	4.43	<0.001
	II	V	6.51	-2.88	9.39	<0.001
	III	II	4.69	6.51	-1.82	<0.001
	III	IV	4.69	2.08	2.61	<0.001
	III	V	4.69	-2.88	7.57	<0.001
	V	IV	-2.88	2.08	-4.96	<0.001
Main Operating profit margin	I	II	46.38	34.28	12.1	<0.001
	I	III	46.38	24.07	22.31	<0.001
	I	IV	46.38	19.175	27.205	<0.001
	I	V	46.38	13.615	32.765	<0.001
	II	IV	34.28	19.175	15.105	<0.001
	II	V	34.28	13.615	20.665	<0.001
	III	II	24.07	34.28	-10.21	<0.001
	III	IV	24.07	19.175	4.895	<0.001
	III	V	24.07	13.615	10.455	<0.001
	V	IV	13.615	19.175	-5.56	<0.001
Cost-Expense-Profit Margin	I	II	36.33	18.23	18.1	<0.001
	I	III	36.33	9.67	26.66	<0.001
	I	IV	36.33	4.415	31.915	<0.001
	I	V	36.33	-8.725	45.055	<0.001
	II	IV	18.23	4.415	13.815	<0.001
	II	V	18.23	-8.725	26.955	<0.001
	III	II	9.67	18.23	-8.56	<0.001
	III	IV	9.67	4.415	5.255	<0.001
	III	V	9.67	-8.725	18.395	<0.001
	V	IV	-8.725	4.415	-13.14	<0.001
Operating Profit Ratio	I	II	25.79	14.64	11.15	<0.001
	I	III	25.79	8.53	17.26	<0.001
	I	IV	25.79	4.175	21.615	<0.001
	I	V	25.79	-7.915	33.705	<0.001
	II	IV	14.64	4.175	10.465	<0.001
	II	V	14.64	-7.915	22.555	<0.001
	III	II	8.53	14.64	-6.11	<0.001
	III	IV	8.53	4.175	4.355	<0.001
	III	V	8.53	-7.915	16.445	<0.001
	V	IV	-7.915	4.175	-12.09	<0.001

TABLE 7. ***(Continued.)*** **Results of Dunn's test.**

ROA	I	II	15.29	17.27	-1.98	0.006
	I	III	15.29	15.26	0.03	0.63
	I	IV	15.29	10.46	4.83	<0.001
	I	V	15.29	1.85	13.44	<0.001
	II	IV	17.27	10.46	6.81	<0.001
	II	V	17.27	1.85	15.42	<0.001
	III	II	15.26	17.27	-2.01	<0.001
	III	IV	15.26	10.46	4.8	<0.001
	III	V	15.26	1.85	13.41	<0.001
	V	IV	1.85	10.46	-8.61	<0.001
Net Profit Margin	I	II	22.352	12.845	9.507	<0.001
	I	III	22.352	7.599	14.754	<0.001
	I	IV	22.352	3.612	18.741	<0.001
	I	V	22.352	-9.921	32.273	<0.001
	II	IV	12.845	3.612	9.234	<0.001
	II	V	12.845	-9.921	22.766	<0.001
	III	II	7.599	12.845	-5.247	<0.001
	III	IV	7.599	3.612	3.987	<0.001
	III	V	7.599	-9.921	17.519	<0.001
	V	IV	-9.921	3.612	-13.532	<0.001
Net Profit Margin on Total Assets	I	II	9.542	7.17	2.373	<0.001
	I	III	9.542	5.039	4.504	<0.001
	I	IV	9.542	2.203	7.339	<0.001
	I	V	9.542	-2.936	12.478	<0.001
	II	IV	7.17	2.203	4.967	<0.001
	II	V	7.17	-2.936	10.106	<0.001
	III	II	5.039	7.17	-2.131	<0.001
	III	IV	5.039	2.203	2.836	<0.001
	III	V	5.039	-2.936	7.975	<0.001
	V	IV	-2.936	2.203	-5.139	<0.001
Net Profit Growth Rate	I	II	17.903	12.901	5.001	0.034
	I	III	17.903	14.508	3.394	0.046
	I	IV	17.903	12.005	5.898	<0.001
	I	V	17.903	-28.43	46.333	<0.001
	II	IV	12.901	12.005	0.896	0.004
	II	V	12.901	-28.43	41.331	<0.001
	III	II	14.508	12.901	1.607	0.779
	III	IV	14.508	12.005	2.503	0.001
	III	V	14.508	-28.43	42.938	<0.001
	V	IV	-28.43	12.005	-40.435	<0.001
Accounts Receivable Turnover Ratio	I	II	6.29	5.61	0.68	0.045
	I	III	6.29	5.22	1.07	0.017
	I	IV	6.29	4.945	1.345	<0.001
	I	V	6.29	4.35	1.94	<0.001
	II	IV	5.61	4.945	0.665	0.001
	II	V	5.61	4.35	1.26	0.014
	III	II	5.22	5.61	-0.39	0.569
	III	IV	5.22	4.945	0.275	0.005
	III	V	5.22	4.35	0.87	0.029
	V	IV	4.35	4.945	-0.595	0.66
Cash Flow to Sales Ratio	I	II	0.19	0.12	0.07	<0.001
	I	III	0.19	0.07	0.12	<0.001
	I	IV	0.19	0.05	0.14	<0.001
	I	V	0.19	0.04	0.15	<0.001
	II	IV	0.12	0.05	0.07	<0.001
	II	V	0.12	0.04	0.08	<0.001
	III	II	0.07	0.12	-0.05	<0.001
	III	IV	0.07	0.05	0.02	0.006
	III	V	0.07	0.04	0.03	0.003
	V	IV	0.04	0.05	-0.01	0.225

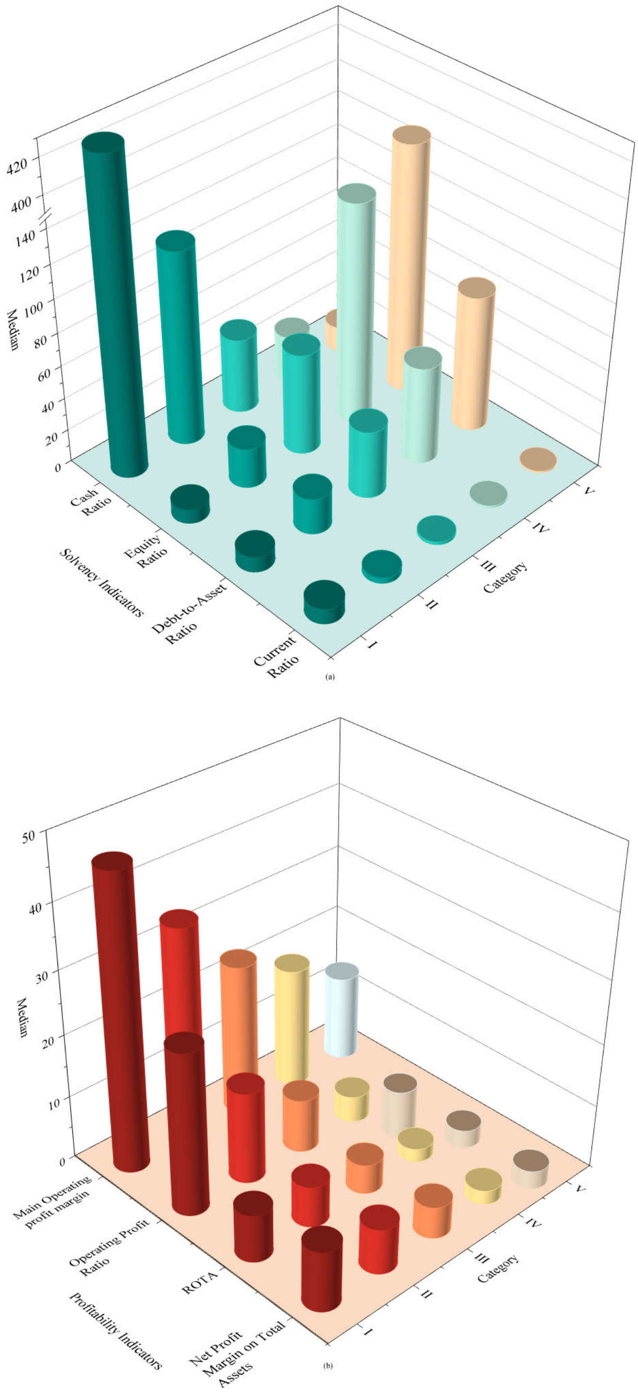

FIGURE 7. **Average trend graph of key financial indicators affecting performance rating.**

Fig. 7 provides an overview of the comparative trends in key financial indicators for each category.

Category "I" enterprises exhibit notable distinctions in terms of profitability, characterized by their high performance in profitability indicators such as Main Operating Profit Margin, Net Profit Margin, and Net Profit Margin on Total Assets compared to other categories. This suggests that Category "I" enterprises effectively manage costs and expenses while delivering products or services, resulting in superior profitability. Additionally, Category "I" enterprises demonstrate the highest Total Asset Growth Rate, indicating their enhanced market competitiveness and growth potential. Furthermore, Category "I" enterprises generally exhibit favorable financial indicators such as the Current Ratio, Cash Ratio, and Equity Ratio, surpassing other categories and signifying a balanced long, medium, and short-term liability structure. A higher Cash Flow to Sales Ratio signifies the ability of a company to maintain adequate liquidity for sustained business development based on sales revenue. In conclusion, Category "I" enterprises display heightened profitability, lower-risk debt, substantial growth prospects, and exceptional operational capacity, thereby indicating minimal financial risk.

Category "V" enterprises exhibit a higher prevalence of problematic financial indicators, particularly in relation to profitability and solvency. The Current Ratio and Cash Ratio of Category "V" enterprises are lower than those of other enterprises in the same industry, while the Debt Asset Ratio is the highest of all categories, which may mean that the enterprise is incapable of effectively repaying its debts in the short term or of mobilizing cash assets quickly in an emergency. In terms of profitability, the Net Profit Margin and ROTA of Category "V" companies are mostly negative or much lower than those of other companies in the same industry, which means that the company is unable to make sufficient profits with the same sales revenue.

The financial performance of the remaining three categories of enterprises falls within the spectrum between categories "I" and "V". Category "II" enterprises exhibit significantly higher ROA, Accounts Receivable Turnover Ratio, and Cash Flow to Sales Ratio compared to the other two categories. These findings indicate that while category "II" companies may not be the most profitable, they demonstrate favorable development trends and efficient utilization of assets. The higher Accounts Receivable Turnover Ratio and Inventory Turnover Ratio reflect improved capital utilization efficiency, resulting in reduced capital costs and operational risks. A smaller proportion of Accounts Receivable indicates stronger bargaining power over downstream customers, successful product sales to some extent, and enhanced profitability stability. A high Cash Flow to Sales Ratio indicates that the company is effectively managing its working capital and operating efficiently. It may indicate that the company has a healthy cash flow and is less reliant on external financing to fund its operations.

The examination of Table 6, Table 7 and Fig. 7 reveals a substantial disparity in the level of financial risk across the various categories. This observation provides evidence supporting the rationality and reliability of utilizing the K-Means method to classify the level of financial risk among diverse companies.

C. RISK FORECASTING OF NGBOOST

To increase the model's resilience, the unbalanced sample distribution of firms' financial risk levels obtained from

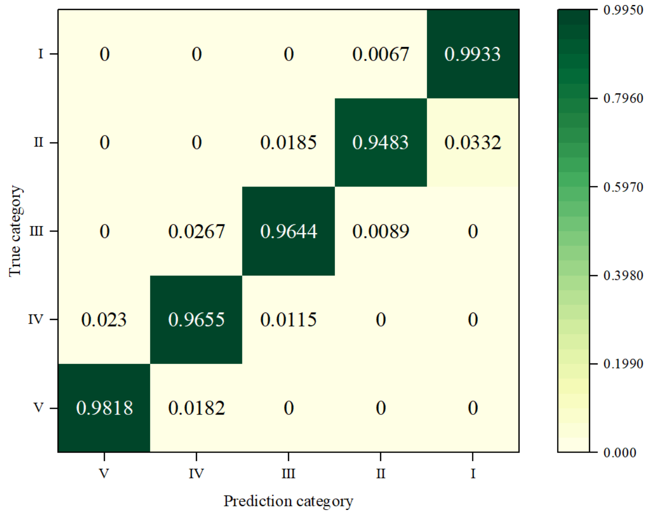

FIGURE 8. SMOTE-ENN-NGBoost confusion matrix.

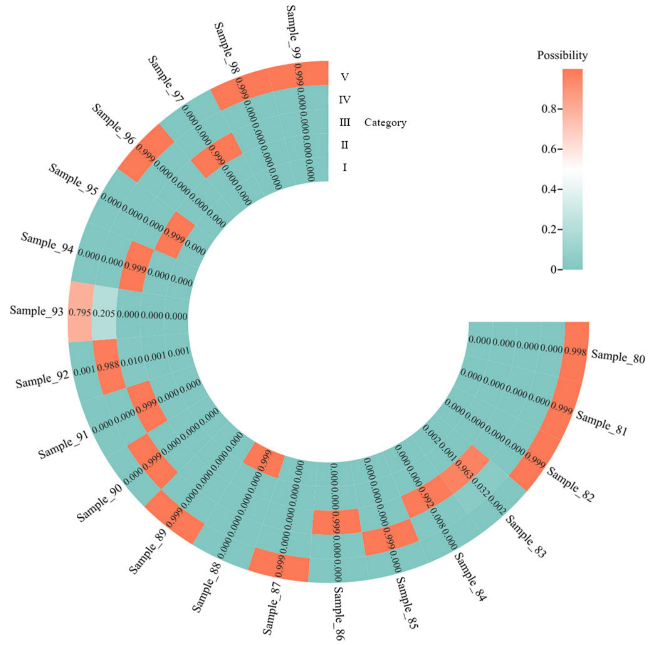

FIGURE 9. Probability prediction results for selected samples.

K-means clustering was addressed through data enhancement. The combined SMOTE-ENN sampling method was employed, resulting in a total of 6923 samples, with approximately 1385 samples for each category. This controlled and balanced distribution of samples significantly improved the stability of the model.

The NGBoost model was trained using the scikit-learn framework. The decision tree was chosen as the model's base learner, and after continuous experiments and parameter adjustments, the criterion of the decision tree was set to "friedman_mse", and the max depth of a single decision tree was set to 8 to avoid overfitting. The "Score" of the model was set to "LogScore", the "n_estimators" was 224, the learning rate was set to 0.0237, the number of data in the training and test sets was set to a ratio of 4:1, with 5538 training data and 1385 test data randomly selected for training, and the final accuracy of the model was calculated. The parameter settings are shown in Table 8.

The model's performance validation is carried out using the test dataset, and the accuracy confusion matrix is presented in Fig. 8. The results indicate high recognition rates for the "I" and "V" categories, with accuracies of 99.33% and 98.18% respectively. This suggests that the model is effective in identifying companies with very low and very high financial risk. For the "II", "III" and "IV" categories, the recognition rates were 94.83%, 96.44% and 96.55% respectively. The model's average accuracy was determined to be 97.18%, showing that the suggested model is both reliable and accurate.

NGBoost model predicts the likelihood of a sample falling into a certain category, which holds significance for corporate financial risk management and investors. Fig. 9 illustrates the probabilities of the samples 80-99 belonging to categories "I" to "V". Taking sample_93 as an example, it exhibits a 79.5% probability of falling into category "V", indicating a higher level of uncertainty compared to other companies. This implies that the sample may excel in profitability but lag in terms of cost control and debt management. To mitigate potential risks, investors should conduct a thorough analysis and comparison of the firm's financial position. For business managers, this probability enables a better understanding of their overall financial situation, industry position, and brand influence, facilitating well-informed decision-making for future directions.

D. COMPARATIVE SIMULATION EXPERIMENTS

To further substantiate the rationality and superiority of suggested model, on the basis of CRITIC-TOPSIS and K-means clustering, a comparison was conducted with various models, including the NGBoost model without data augmentation, the SMOTE-ENN-CNN model, the CNN model without data augmentation, the XGBoost model, the LightGBM model, the SVM, and the KNN model. Four multi-categorical metrics, namely accuracy, F1 score, recall, and precision, were utilized to assess the performance of each model. Fig. 10 presents the performance of each model, and the specific values are detailed in Table 9.

Fig. 10 shows how the suggested model in this study beats the other models in terms of accuracy, F1 score, recall, and precision. The corresponding values in Table 9 support this observation, as the proposed model achieved an accuracy of 97.18%, F1 score of 97.08%, precision of 97.12%, and recall of 97.07%. These metrics were superior by 6.72%, 7.56%, 6.45%, and 8.51%, respectively, compared to the model without data enhancement using the SMOTE-ENN algorithm. The Convolutional Neural Network (CNN) model ranked as the second most effective, attaining accuracy of 93.43%, F1 score of 93.42%, precision of 93.87%, and recall of 93.43%. These metrics were better by 6.08%, 6.14%, 5.66%, and 6.08%,

TABLE 8. Model parameter.

Boosting iterations	Learning rate	Dist	Scoring rule	Criterion	Max depth
224	0.0237	k_categorical	LogScore	friedman_mse	8

TABLE 9. Comparison of accuracy rates of different models.

Model	Accuracy	F1	Precision	Recall
NGBoost(SMOTE-ENN)	0.9718	0.9708	0.9712	0.9707
NGBoost	0.9046	0.8952	0.9067	0.8856
CNN(SMOTE-ENN)	0.9343	0.9342	0.9387	0.9343
CNN	0.8735	0.8728	0.8821	0.8735
XGBoost	0.8679	0.8609	0.8694	0.8544
LightGBM	0.8557	0.8348	0.8718	0.8088
SVM	0.8357	0.6984	0.8962	0.6625
KNN	0.828	0.7433	0.8915	0.698

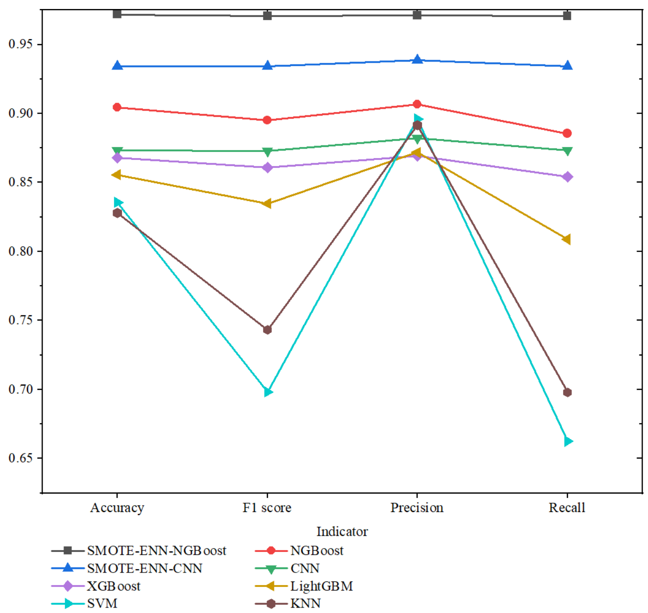

FIGURE 10. Models performance comprehensive comparison.

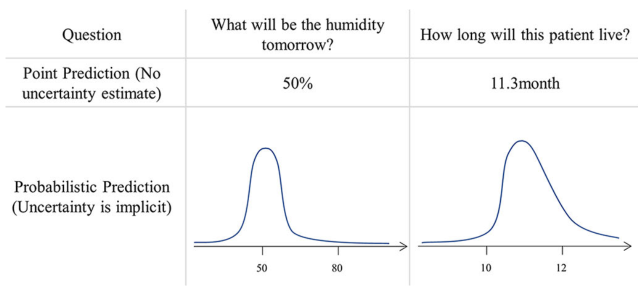

FIGURE 11. The difference between point prediction and probabilistic prediction.

respectively, compared to the model without data augmentation. Furthermore, the proposed model exhibited significantly superior performance compared to the commonly used SVM and KNN models across various evaluation metrics. Additionally, the proposed model also outperformed the XGBoost and LightGBM models, which share similar principles.

In the realm of finance, where decisions can carry significant financial implications and risks, it is crucial to have reliable and interpretable models that can guide decision-making. In contrast, neural networks are often regarded as ''black box'' models, lacking transparency in their decision-making process. Moreover, neural network models necessitate high computational power, which inevitably adds to the enterprise's costs. The Ensemble Learning algorithm employed in this study exhibits low sample dependency and high accuracy, making it ideal for aiding companies in accurately predicting future growth while keeping costs in check. While the SVM and KNN algorithms offer good interpretability, they do not perform as well as decision trees and Ensemble Learning algorithms when confronted with complex data.

One of the significant advantages of NGBoost is its ability to provide probabilistic predictions, which is of crucial importance in the field of risk assessment and financial analysis. Whereas in algorithms like XGBoost and LightGBM, only point predictions can be provided. The difference between point prediction and probabilistic prediction is demonstrated in Fig. 11. Financial markets are inherently uncertain and volatile. Probabilistic predictions provide a measure of uncertainty by offering a distribution of possible outcomes rather than a single point estimate. This is critical for understanding the range of potential risks associated with a financial decision or portfolio. Furthermore, financial institutions and investors often need to manage risks effectively. Probabilistic predictions allow them to assess not only the expected return but also the likelihood of different outcomes, including worst-case scenarios. This information helps in making informed decisions about asset allocation and risk mitigation strategies.

NGBoost's key innovation is in employing the natural gradient to perform gradient boosting by casting it as a problem of determining the parameters of a probability distribution. In the realm of machine learning, the efficacy of traditional gradient-based optimization techniques is often hindered when it comes to learning multi-parameter

TABLE 10. PMF of categorical distribution.

PMF	Value
P(X=Ⅰ)	0.000189
P(X=Ⅱ)	0.00015927
P(X=Ⅲ)	0.00015159
P(X=Ⅳ)	0.00018673
P(X=Ⅴ)	0.99931341

probability distributions, such as the ubiquitous Normal distribution. However, a promising solution emerges in the form of natural gradients, a concept that has revolutionized the training process for probabilistic models. By incorporating natural gradients into the optimization scheme, we enhance the stability and efficiency of model fitting. One notable implementation of this paradigm shift is exemplified in NGBoost. NGBoost not only harnesses the power of natural gradients but also extends its versatility to accommodate classification distributions. Users can conveniently specify these distributions through the NGBClassifier() constructor, leveraging the 'Dist' argument. This study is solving a multi-classification problem, so the prediction result conforms to the Categorical distribution, and the model will return a parameter Probability Mass Function (PMF) describing the distribution of the predicted samples. Taking the first sample as an example, the returned parameter PMF is shown in Table 10.

Furthermore, the proposed model combines the SMOTE-ENN algorithm for data augmentation. This integration significantly reduces the subjectivity of business risk evaluation and enhances the model's accuracy.

IV. CONCLUSION

This paper utilized the annual financial statement data of 4,502 A-share companies for the year 2021. The CRITIC-TOPSIS approach was used for evaluating the financial performance of these organizations in depth. Subsequently, the comprehensive evaluation results were categorized into five financial risk levels through K-means clustering, which were further utilized as the sample labels in the analysis. To achieve intelligent prediction of financial risk levels, the NGBoost classification model was employed. Additionally, to address the issue of sample imbalance commonly observed in similar studies, data augmentation was performed on the clustering results using combined sampling SMOTE-ENN, consequently, the model's prediction accuracy is improved. The following are the study's key findings:

(1) This study introduces the CRITIC-TOPSIS fusion model as a novel approach to comprehensively assess the financial risk of firms. By combining the objective assignment method with rigorous multidimensional ranking, this model offers a more scientific and reasonable approach compared to using a single evaluation method. It provides decision-makers with a scientific basis for evaluation. The highest score of 0.464 is achieved by *Sino Biological, Inc.*, while the lowest score of 0.283 is obtained by ∗ST *Inner Mongolia Western Resources Co., Ltd*. The average score across all companies is 0.368. Among the indicators considered, Debt-to-Asset Ratio, Fixed Asset Ratio, and Operating Revenue have a significant impact on a corporation's financial position, while the Cash Flow to Sales Ratio has a relatively lesser impact.

(2) This paper employs the K-means algorithm to categorize a company's financial risk into five distinct levels, denoted as "I", "II", "III", "IV", and "V". These categories represent very low financial risk, low financial risk, average financial risk, light financial warning, and heavy financial warning, respectively. This classification approach provides more detailed and informative insights compared to a binary classification of simply labeling companies as "ST" or "non ST". The analysis reveals that only 6.55% of the total number of companies fall into category "I", while 6.42% belong to category "V". Notably, "ST" enterprises are predominantly found in categories "IV" and "V", while categories "I" and "II" do not include any "ST" companies.

(3) The model integrating SMOTE-ENN and NGBoost proposed in this paper achieved a classification accuracy of 97.18%, surpassing other models, including Convolutional Neural Network models, while also reducing the computational requirements. The accuracy improvement of 6.72% compared to the model without data enhancement using the SMOTE-ENN algorithm highlights the reliability of the proposed approach.

The model developed in this work has practical implications for corporate financial risk early warning and provides insights into comprehensive evaluation challenges. Future research could explore the inclusion of non-financial indicators in the financial risk evaluation index system to provide a more comprehensive analysis.

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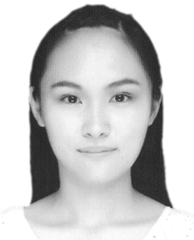

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